NUMERICAL INTEGRATION

Definite integrals can be approximated using numerical integration methods. Two common methods are Trapezoidal rule and Simpson's rule.

* Numerical methods carry errors, which be computed.

TRAPEZOIDAL RULE

Suppose $f(x)$ is a continuous and differentiable on interval $x \in [a,b]$, then the definite integral of $f(x)$ on $[a,b]$ can be approximated using trapezoidal rule, which states:

$$\int_a^b f(x)\,dx \approx A_n^T = \frac{S}{n}\left[f(a) + 2f(x_1) + 2f(x_2) + 2f(x_3) + \dots + 2f(x_{n-1}) + f(b)\right]$$

$$A_n^T = \frac{S}{n}\left[F + 2T + L\right]$$

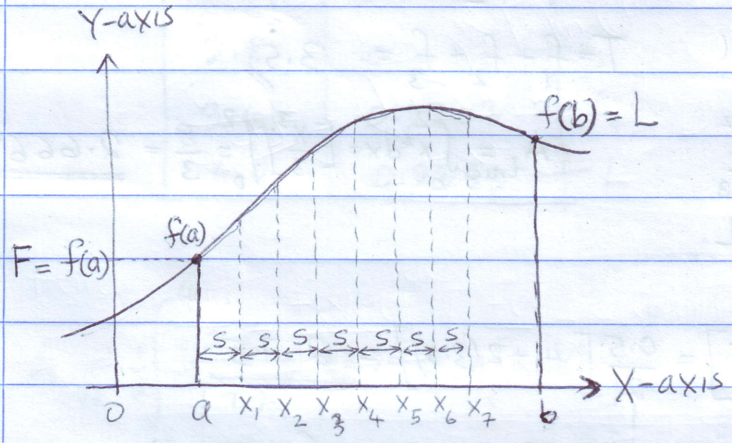
, where $A = f(a) \leftarrow$ First ordinate

$L = f(b) \leftarrow$ Last ordinate

$T = f(x_1) + f(x_2) + \dots + f(x_{n-1})$

$S = \Delta x = \frac{b-a}{n} \leftarrow$ Interval size

$x_i = a + is$, $i = 1, 2, 3, \dots, n-1$



ERROR OF TRAPEZOIDAL RULE

If $f(x)$ is continuous and twice differentiable on $[a,b]$, then error of trapezoidal rule is

$$|E_n^T| = \left|\int_a^b f(x)\,dx - A_n^T\right| \leq \frac{(b-a)^3}{12n^2} M, \quad \text{where } M = \max_{x \in [a,b]} |f''(x)|$$

Percentage Error:

$$\frac{|A_{true} - A_n^T|}{A_{true}} \times 100\%$$

Where: $A_{true} = \int_a^b f(x)\,dx \leftarrow$ exact value of integral.

$$\frac{|E_n^T|}{A_{true}} \times 100\%$$

Example 1

Use trapezoidal rule to approximate the integral

$$\int_0^2 x^2\,dx$$

(a) Using $n=4$ and $n=8$ intervals.

(b) Find the percentage error of the method for $n=4$ and $n=8$.

Solution

(a) Here $a=0$, $b=2$, $n=4$ $\quad S=\Delta x=\frac{b-a}{n}=\frac{2-0}{4}=0.5$ (step size)

Iteration Table:

$x_i = a+iS = 0+0.5i = 0.5i,\ i=1,2,\ldots$

$f(x_i)=x_i^2$

x_i	$f(x_i)$	
0	0	← F
0.5	0.25	← f_1
1.0	1.0	← f_2
1.5	2.25	← f_3
2.0	4	← L

$F+L = 0+4 = 4$

$T = f_1+f_2+f_3 = 3.5$

$* A_{true} = \int_0^2 x^2 dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3} = 2.6667$

$$A_4^T \approx \frac{S}{2}\left[F+L+2T\right] = \frac{0.5}{2}\left[4+2(3.5)\right] = 2.75$$

(b) $a=0,\ b=2,\ n=8,$ $\quad S=\Delta x=\frac{b-a}{n}=\frac{2-0}{8}=0.25$ (step size)

$x_i = 0.25i,\ i=1,2,3,\ldots$

$f(x_i)=x_i^2$

x_i	$f(x_i)$	
0	0	F
0.25	0.0625	f_1
0.5	0.25	f_2
0.75	0.5625	f_3
1.0	1.0	f_4
1.25	1.5625	f_5
1.5	2.25	f_6
1.75	3.0625	f_7
2.0	4	L

$F+L = 4$

$T = f_1+f_2+\ldots+f_7 = 8.75$

$$A_8^T = \frac{S}{2}\left[F+L+2T\right] = \frac{0.25}{2}\left[4+2(8.75)\right] = 2.6875$$

(b) $E_4^T = |2.6667-2.75| = 0.0833$ $(n=4)$ 3.12%

$E_8^T = |2.6667-2.6875| = 0.0208$ $(n=8)$ 0.78%

$\% E_8^T = \frac{0.0208}{2.6667} \times 100\% = 0.78\%$

Example 2

Given a definite integral $\int_1^4 \frac{x}{1+x^2}\,dx$

(a) Use trapezoidal rule of $n=5$ intervals to approximate the integral.

(b) Find the true value, A_{true} of the integral.

(c) Find the error and percentage error of the method at $n=5$.

Solution

(a) $a=1,\ b=4,\ n=5$ then $S=\Delta x=\frac{b-a}{n}=\frac{4-1}{5}=\frac{3}{5}=0.6$ (step size)

$x_i=a+is=1+0.6i,\ i=1,2,\ldots$ $\quad f(x_i)=\frac{x_i}{1+x_i^2}$ Correct $f(x_i)$ to 4 decimal places.

Table:

x_i	$f(x_i)$	
1	0.5	F
1.6	0.4494	f_1
2.2	0.3767	f_2
2.8	0.3167	f_3
3.4	0.2707	f_4
4.0	0.2353	L

$F+L=0.5+0.2353=0.7353$

$T=f_1+f_2+f_3+f_4=1.4135$

$A_5^T=\frac{S}{2}\left[F+L+2T\right]$

$=\frac{0.6}{2}\left[0.7353+2(1.4135)\right]$

$\therefore A_5^T=1.0687$

(b) $A_{true}=\frac{1}{2}\int_1^4 \frac{2x}{1+x^2}\,dx=\frac{1}{2}\ln\left[1+x^2\right]\Big|_1^4$

$=\frac{1}{2}\left[\ln 17-\ln 2\right]=1.0700$

$\therefore A_{true}=1.0700$

(c) Error: $E_5^T=\left|A_{true}-A_5^T\right|=\left|1.0700-1.0687\right|=0.0013$
$(n=5)$

Percentage Error: $\frac{\left|A_{true}-A_5^T\right|}{A_{true}}\times 100\%=\frac{0.0013}{1.0700}\times 100\%=0.1215\%$

*Conclusion: "As n increase the error decrease"

Example 3

Use trapezoidal rule to approximate the integral

$$\int_1^2 \ln(1+x)\,dx \quad \text{with a maximum error of } 0.001.$$

Solution

Max error of Trapezoidal rule $\left|E_n^T\right| \leq \frac{(b-a)^3}{12n^2}|M|$, where $M = \max\limits_{x\in[a,b]} |f''(x)|$.

Here $a=1$, $b=2$, $f(x) = \ln(1+x)$, $f'(x) = \frac{1}{1+x}$, $f''(x) = \frac{-1}{(1+x)^2}$

Since $f(x)$ and $f''(x)$ are increasing on $[1,2]$, then $|M| = |f''(2)| = \frac{1}{9}$

$$\left|E_n^T\right| \leq \frac{(2-1)^3}{12n^2}\cdot\left(\frac{1}{9}\right) \Rightarrow \frac{1}{108n^2} \leq 0.001$$

We need $n^2 \geq \frac{1000}{108} = 9.26 \Rightarrow n \geq \sqrt{9.26} \approx 3.04$, So that choose: $n = 4$

$$S = \Delta x = \frac{b-a}{n} = \frac{2-1}{4} = 0.25$$

$f(x_i) = \ln(1+x_i)$ $\quad x_i = 1+0.25i,\ i=1,2,3,4$

Table:

Note: Always correct $f(x_i)$ value to 4 d.p.

x_i	$f(x_i)$	
1	0.6931	F
1.25	0.8109	f_1
1.5	0.9163	f_2
1.75	1.0116	f_3
2.0	1.0986	L

$F + L = 1.7917$

$T = f_1 + f_2 + f_3 = 2.7388$

$$A_4^T = \frac{S}{2}\left[F + L + 2T\right] = \frac{0.25}{2}\left[1.7917 + 2(2.7388)\right]$$

$$\therefore A_4^T = 0.9087$$